UML Architecture Package
Physiological Digital Twin Berbasis CAPAR

Dokumen ini mendefinisikan representasi UML tingkat arsitektur untuk sistem Physiological Digital Twin CAPAR. Package terdiri dari Use Case Diagram, Component Diagram, Sequence Diagram, dan Deployment Diagram dalam bentuk spesifikasi tekstual yang dapat diterjemahkan menjadi model UML.

1. Use Case Diagram

Actors:

1. Participant
- memakai wearable
- memberikan EMA/context
- melihat hasil personal

2. Clinician/Researcher
- melihat resilience status
- mengevaluasi trajectory
- membaca explanation

3. Concept Engineer
- mengatur physiological model
- mengatur rule dan threshold

4. Software Engineer
- melakukan deployment
- maintenance sistem

Main Use Cases:

UC-01 Acquire Physiological Data
UC-02 Manage Context Input
UC-03 Integrate Clinical Information
UC-04 Process Physiological Signal
UC-05 Extract CAPAR Features
UC-06 Estimate Resilience State
UC-07 Generate CAPAR Event-State
UC-08 Display Recovery Trajectory
UC-09 Generate XAI Explanation
UC-10 Export Research Report

2. Component Diagram

```
+----------------------------------------------+
| Application Layer                            |
| Dashboard | XAI Viewer | Report Generator      |
+----------------------------------------------+
                    |
+----------------------------------------------+
| Physiological Digital Twin Layer             |
| Resilience Engine                            |
| State Estimator                              |
| CAPAR Finite State Machine                   |
| Phenotype Classifier                         |
+----------------------------------------------+
                    |
+----------------------------------------------+
| Evidence Processing Layer                    |
| Feature Extraction                           |
| Personal Baseline Engine                     |
| Context Engine                               |
| Quality Gate                                 |
+----------------------------------------------+
                    |
+----------------------------------------------+
| Acquisition Layer                            |
| Wearable | Clinical DB | EMA                 |
+----------------------------------------------+
```

3. Component Responsibility

- Data Acquisition Component: menerima data sensor dan clinical input.
- Quality Component: melakukan validasi dan confidence scoring.
- Feature Component: menghitung HR, HRV, DFA, recovery, relapse feature.
- Baseline Component: membuat baseline personal-contextual.
- CAPAR Engine: melakukan deviation, persistence, recovery, relapse detection.
- Digital Twin Engine: memperkirakan resilience state.
- XAI Engine: menyediakan evidence dan uncertainty.

4. Sequence Diagram: Continuous Monitoring

```
Participant
   |
   | wearable signal
   v
Data Acquisition Service
```

```
   |
   | raw data
   v
Preprocessing Service
   |
   | clean signal
   v
Feature Engine
   |
   | features
   v
CAPAR Engine
   |
   | event-state
   v
Digital Twin Engine
   |
   | resilience status
   v
XAI Service
   |
   v
Dashboard
```

5. Sequence Logic CAPAR Engine

Input:
u(k), d(k), c(k), y(k)

Process:

1. Validate observation.
2. Determine context.
3. Compare with personal baseline.
4. Calculate deviation score.
5. Evaluate persistence.
6. Detect recovery.
7. Detect relapse.
8. Update state.

Output:

s(k):
Baseline Mature
No Deviation
Deviation

Persistent Deviation
Recovery Start
Full Recovery
Relapse

6. Deployment Diagram

```
                  User Device
                 Mobile/Web UI
                      |
                      |
                API Gateway
                      |
        --------------------------------
        |                         |
     Cloud Layer                Edge Layer
        |                         |
 Database Server            Wearable Gateway
        |
 Model Inference Server
        |
 CAPAR Engine
        |
 Digital Twin Engine
```

7. Traceability Architecture

Input:
Wearable + Clinical + Context

↓

Feature:

HR, HRV, DFA, recovery metrics

↓

Latent Variable:

Autonomic tone
Cardiac reserve
Recovery dynamics

↓

Resilience Domain:

Vulnerability
Cardiac Reserve
Autonomic Reserve
Recovery Capacity
Regulation Stability

↓

Output:

Risk
Trajectory
Phenotype
Decision Support
XAI

8. Architectural Validation

Validasi UML dilakukan dengan memastikan setiap requirement SRS memiliki representasi pada komponen arsitektur dan setiap komponen memiliki tanggung jawab yang jelas.